

PARTS LIST

Semiconductors:

IC1	- 7805, 5V regulator
IC2, IC3	- NE555 timer
IC4	- MOC3021 optocoupler
SCR1	- 2P4M SCR
TRIAC1	- BT136 triac
T1, T2	- 2N2222 npn transistor
BR1	- 2A bridge rectifier
D1-D3	- 1N4007 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R4, R10	- 220-ohm
R2, R3, R5-R8	- 1-kilo-ohm
R9	- 10-kilo-ohm
R11	- 470-ohm
R12	- 39-ohm
VR1	- 1-kilo-ohm potentiometer

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2	- 0.1 μ F ceramic disk
C3	- 0.01 μ F ceramic disk
C4	- 10 μ F, 16V electrolytic
C5	- 0.01 μ F, 400V polyester

Miscellaneous:

X1	- 230V primary to 12V, 2A secondary transformer
M1	- 12V DC motor
M2	- 220V, 12W AC motor

through selection switch S1.

IC3 is also configured as an inverter. The pulse available at its pin 2 is inverted and output at pin 3. It drives the DC motor with the help of SCR1; the output of IC3 is connected to the gate of SCR1. The DC motor (M1) is connected between the rectified output and the anode of SCR1 along with free-wheeling diode D3. The cathode of SCR1 is connected to ground.

The output of IC2 is also connected to pin 1 of optocoupler MOC3021 through current-limiting resistor R10. Pin 2 of the optocoupler is grounded. Pin 6 of IC4 is connected to MT2 terminal of TRIAC1 (BT136) through resistor R11 and pin 4 drives the gate of TRIAC1. A snubber circuit comprising resistor R12 and capacitor C5 connected across TRIAC1 limits the rate of change of voltage.

Operation

The operation of the circuit can be better understood from the waveforms shown in Fig. 2. Point 'A' shows the stepped-down AC waveform and point 'B' the rectified waveform.

When the input voltage at the base

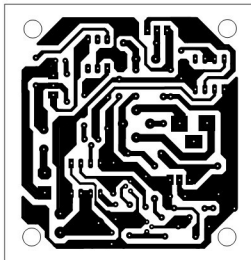


Fig. 3: An actual-size, single-side PCB for the speed controller for AC and DC motors

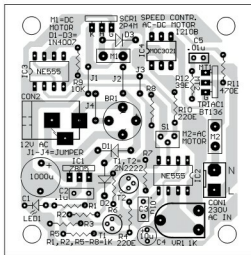


Fig. 4: Component layout for the PCB

of transistor T1 is lower than 0.6V, the transistor is cut off. Transistor T1 produces a very short-duration positive pulse at point 'C' and a negative pulse of the same width at point 'D'. This pulse is used for switching operation.

The monostable multivibrator (built around IC2) triggers when pin 2 of IC2 receives the negative pulse from point 'D' and generates a positive pulse through pin 3 of IC2. The time period of this pulse is determined by potentiometer VR1 and capacitor C4. The pulse width can be varied up to 10 ms by adjusting potentiometer VR1. The positive pulse is shown at point 'E' of Fig. 2.

When the output from pin 3 of IC2 is fed to pin 2 of IC3 through switch S1, IC3 outputs an inverted pulse at its pin 3. The inverted pulse from IC3 controls the gate of SCR1, which controls the firing of SCR1. So the DC motor gets rectified pulsating DC for the positive pulse period of IC3 (as shown in Fig. 2). As the average DC voltage increases, the motor runs faster, and vice versa.

Switch S1 connects the output from pin 3 of IC2 to pin 1 of the optocoupler (IC4) through current-limiting resistor R10. When the output of IC2 goes high, the current passes through the internal LED of the optocoupler and triggers the internal diac of the optocoupler. Simultaneously, the diac provides the gate pulse to TRIAC1 and TRIAC1 is fired for the high pulse period (as shown in Fig. 2) to drive the AC motor. The speed of the motor increases as more voltage is applied, and vice-versa.

Construction

An actual-size, single-side PCB for speed control of AC and DC motors is shown in Fig. 3 and its component layout in Fig. 4.

Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error. First, check the waveform at different points (refer Fig. 2). After checking the waveforms at points 'A' through 'F', connect the DC and AC motors. Using switch S1, select between AC and DC motor speed control.

Caution. Since switching is accomplished by triac, don't touch the circuit parts while AC supply is plugged in. Further, ensure that Live and Neutral lines are not interchanged. ●

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